

The first two years



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Outline

Physical development

- Reflexes
- Growth and proportions
- Sensory development
- Brain development
- Motor development



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Preterm and low birthweight infants

- Weigh less than 2.5 kgs
- May or may not be premature (≤ 37 weeks' gestation)
- Birthweight best available predictor of healthy development
- Babies who are "small-for-dates" are more likely to require support



Photo: Amillia Taylor, born 24/10/06, Florida.

- gestation 22 weeks
- length 24.13 cms
- weight 284 grams

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Body growth

- Very rapid development
- Grow in spurts, gain "baby fat" until about 9 months
- Growth follows the cephalocaudal and proximodistal trends



2 months

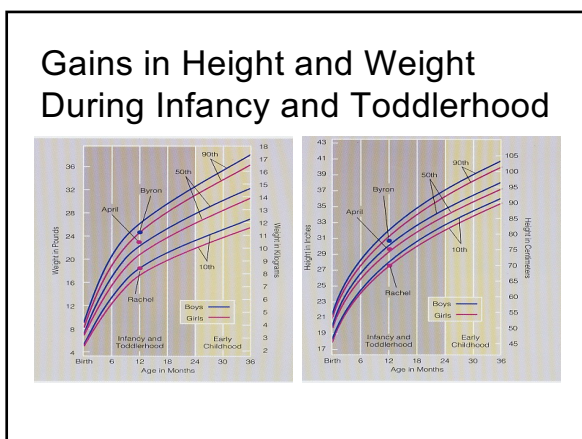


12 months



19 months

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Newborn reflexes

- Inborn, automatic responses to external stimulation
 - Many weaken and disappear after 3-6 months
 - Some survival reflexes remain permanently, eg. blinking

Moro reflex



Babinski reflex



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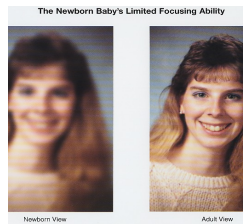
Development of the Senses

- Sensitivity to **touch** and **ability to feel pain** present well before birth
- Sense of **taste** before birth
 - prefer sweet-tastes
- Sense of **smell** less developed but show preference
- **Hearing** well developed before birth

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Vision

- Initial focus and visual acuity limited
- 4-5 months, recognise their mother's face in a photo.
- By 2, visual acuity near adult level.

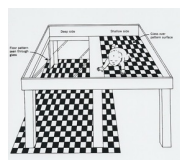


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Perception

- Depth perception
 - Crawling infants distinguish deep and shallow surfaces
 - Crawling promotes a new level of brain organization.

Activity – go to p. 96-97 of textbook. Read about this experiment and describe it to the person next to you.

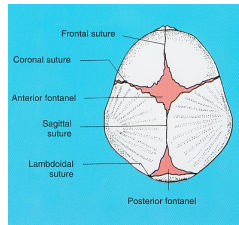


Visual cliff experiment

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Skull growth and brain development

- Fontanelles allow rapid brain growth in first 2 years
- Newborn brain about 25% of adult brain
- By 24 months, 75% of the weight of the adult brain



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Brain development

- Production of **neurons** is largely complete prenatally
- During the first 2 years, huge increase in the **synapses** between them
 - Synaptic pruning
- Rapid increase in **myelination**
 - coat neural fibres with a fatty sheath
 - improves efficiency of information transfer
- **Lateralisation** begins – specialisation of function in either hemisphere

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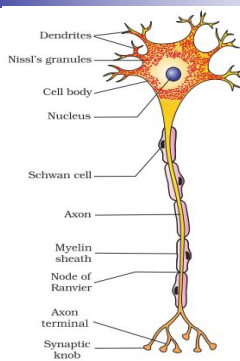


Figure 21.1 Structure of a neuron

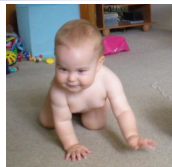
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Nutrition

- Specific nutrition needed to fuel the development of the 1st 2 years
- Brain development best supported by breastfeeding
 - Uniquely adapted for human infants, responsive to the infant's needs
 - Protects from illness
 - Promotes emotional responsiveness

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Motor development



- Follows the cephalocaudal and proximodistal trends
- **Gross motor development** – control over movement around the environment
- **Fine motor development** – smaller more precise movements
- Predictable sequence, variable rates of mastery
- Increasingly complex combinations of skills – **dynamic systems perspective**

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Milestones of Reaching and Grasping

- **Prereaching**
- **Reaching**
 - two hands, then one
- **Ulnar Grasp**
 - Adjust grip to object
 - Move objects hand to hand
- **Pincer Grasp**



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Outline

Cognitive development

- Behavioural learning
- Piaget's sensorimotor stage
- Information processing
- Language development

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Behavioural learning

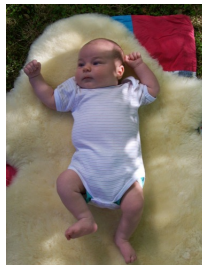
- **Operant conditioning** – act on the environment and this behaviour is followed by a stimulus
 - either increases (reinforces) the behaviour or decreases it through punishment
- **Habituation** – repetition of a stimulus reduces the response. A new stimulus results in recovery of the response – dishabituation.
- **Imitation** – babies can imitate soon after birth
 - Deferred imitation

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Piaget's Cognitive-Developmental Theory

The sensorimotor stage (0 – 2 years)

- Infants learn through their own developing sensory and motor activity
- Shift from reflexive to intentional behaviour



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Reflexive Schemes (birth to 1 month)

- Schemes consist of reflexes: basis of sensorimotor intelligence
- Only a triggering stimulus needed to initiate reflexive action



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1 to 4 months

- Schemes based on attempts to repeat behaviours that first occurred by chance.
- Action and response both centred within the infant's own body.
- Beginnings of voluntary control



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4 to 8 months

- Motor development allows infants to sit up, reach for, manipulate objects
- Turn attention outside own body
- Repeat actions that bring interesting results



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8 to 12 months

- Intentional, goal-directed, behaviour
- Combine schemes to solve problems
- Object permanence developing
 - Make A not B error (seek in first, not the last place an object was hidden)



Object permanence – the understanding that objects still exist even when they cannot be seen

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Piaget - Stage 1 - Sensorimotor stage : Object Permanence



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12 to 18 months

- Repeat actions with variation, exploring through trial and error
- Object permanence achieved (no A not B error)



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18 –24 Months

Symbolic thought

- ☐ Mentally represent objects and events not present using symbols - words, numbers and pictures
- ☐ Solve problems symbolically (not just trial and error).

Mental representation
– internal depictions
of information that the
mind can manipulate



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Information processing

■ Attention

- ☐ Joint attention

■ Memory

- ☐ Implicit memory
- ☐ Explicit memory
- ☐ Infantile amnesia

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Language Acquisition

■ Behaviourist perspective

- ☐ B.F. Skinner: imitation, operant conditioning

■ Nativist perspective

- ☐ Chomsky: innate system for mastering rules of language - Language Acquisition Device (LAD)
- ☐ Sensitive period for language acquisition, especially first 5 years

■ Interactionist perspective

- ☐ Interaction of innate abilities and environmental influences

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Influencing language acquisition



- Adults use child-directed speech
 - Exaggerated expression, high-pitched voice, very clear pronunciation
- Recasting, expanding, labelling
- Language rich environments enhance brain development, intelligence and social skills

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First Words

- Babbling
 - Activated by human speech
 - Becomes focussed on familiar **phonemes** – the sounds of a language
- First words
 - Nominals first – objects, people, events
 - Building understanding of **semantics** (word meanings)
- Toddlers begin linking words – **telegraphic speech**
- Rapidly learn vocabulary through **fast mapping** – linking new words to concepts

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Outline

Socioemotional development

- Psychosocial theory
 - Erikson (trust versus mistrust, autonomy versus shame and doubt)
- Self Development
- Temperament
- Attachment Theory
 - Bowlby
 - Ainsworth

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Erikson's psychosocial theory

Basic trust versus mistrust

Birth to 12-18 mths

- Positive resolution results in the virtue (an ego strength) of **hope**
- Trust is the cornerstone of future identity – the basis for all subsequent stages
- Negative resolution leads to anxiety, difficulty forming intimate relationships

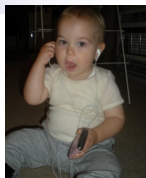


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Autonomy versus shame and doubt

(12-18 mths to 3 yrs)

- Positive resolution needs guidance and appropriate choices
- Beginnings of independent autonomous self – 'I do it!' Virtue of **"will"**.
- Increased sensitivity to others' feelings
- Social referencing



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Development of Basic Emotions

Happiness	■ Smile from birth ■ Social smile 6 to 10 weeks ■ Laugh 3–4 months
Anger	■ General distress from birth ■ Anger 4–6 months
Fear	■ First fears – second half of first year ■ Social Referencing 7-10 months ■ Stranger Anxiety 8–12 months

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Development of self



- Self awareness
- Self recognition
 - By 2, recognise themselves in the mirror and in photos and use 'me' or 'I'
- 18 -30 months, develop a **categorical self**
 - Describe basic characteristic of the self
 - Limited capacity for emotional self-regulation

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Video – Self awareness



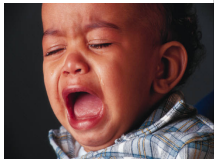
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TEMPERAMENT AND DEVELOPMENT

- Stable individual differences in behavioural style
- Measurable in early infancy, moderated by interaction with the environment

New York Longitudinal Study

- characteristic style of approaching and reacting to people and situations (Thomas, Chess, & Birch, 1968)



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Categories of Temperament

- Easy – 40%
- Difficult – 10%
- Slow-to-warm-up – 15%
- Mixed-pattern– 35%
- **Goodness of fit** is the degree to which parents feel comfortable with their child’s temperament
- Difficult children are less likely to receive sensitive care

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Attachment theory

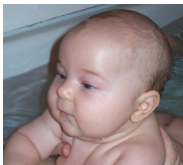
John Bowlby saw attachment as

- A species specific set of behaviours, to maintain proximity and promote species survival
- having a distinct **developmental sequence and a sensitive period during which enduring forms of relating to others develop**

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Bowlby’s Phases of Attachment

Phase 1: birth to 2 months

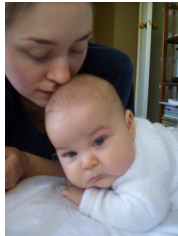


- Indiscriminant social responsiveness
- Reflexive behaviours to gain proximity
- Social smile 5-6 weeks.

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Phase 2: 2 to 7 months

- Discriminant social responsiveness
- Unique relationship formed with primary caregiver
- 4-5 months narrow preferences, still does not protest when they leave
- Social referencing



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Phase 3: 7 to 24 months

Intense attachment to specific caregiver(s)

- Proximity-seeking
 - ☐ Stranger anxiety 6-9 months
 - ☐ Separation anxiety 9-12 months
- Caregiver(s) used as secure base from which to explore.



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Phase 4: 24 months on

- Mental representation of attachment figure
- Separation anxiety decreases
- Over these 4 phases, an enduring **internal working model** of the availability and responsiveness of attachment figures is constructed

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Attachment types

- Ainsworth introduced idea of caregiver as a secure base
- Designed **Strange Situation** to measure quality of attachment relationships (see Stream for video)



- Focused on reunion behaviour

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Strange Situation : Attachment classifications

- **Secure** (65%)
- **Insecure avoidant** (20%)
- **Insecure resistant** (10%)
- **Insecure disorganised** (5 –10%)
- Strong links with adult attachment types
- Intergenerational transmission

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Video: You and me



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